

Research on tool wear state identification method driven by multi-source information fusion and multi-dimension attention mechanism

Peining Wei^a, Rongyi Li^{a,*}, Xianli Liu^a, Haining Gao^b, Mingqiu Dai^c, Yuhan Zhang^a, Wenkai Zhao^a, Erliang Liu^d

^a Key Laboratory of Advanced Manufacturing and Intelligent Technology, Ministry of Education, Harbin University of Science and Technology, Harbin, 150080, China

^b Huanghuai University, Zhumadian 463000, China

^c China Aerospace Science and Technology Corp No 8 Academy-Shanghai Institute of Spacecraft Equipment

^d Changzhou University, Changzhou, 213164, China

ARTICLE INFO

Keywords:

Channel-spatial attention mechanism
Multi-source information fusion
Cutting tool wear state
Deep learning
Time-frequency domain analysis

ABSTRACT

Cutting tool wear condition monitoring technology is the key technology of advanced manufacturing system and a crucial component of machining. The stage of the tool wear has a direct impact on performance of the workpiece and efficacy of machine tool. However, the duplication and lack of wear information and the fuzzy area of the tool wear transition stage are the key factors contributing to the incorrect estimation of the tool wear state when extracting cutting tool wear feature information. As a consequence, this study presents a data fusion from several sources-based intelligent tool wear state detection technique. The fusion of data from several sources can effectively realize the complementarity of machining information. This provides the model with more comprehensive identification data. The mapping between wear state and wear characteristics is precisely established. To address these issues, the attention mechanism of channel and spatial latitude is integrated into the feature extraction. The model that was constructed in the present investigation has a comprehensive identification accuracy of 0.982. The F1 score of initial, normal and severe wear stage of tool wear are 0.977, 0.968 and 0.993, which are better than other models. Experiments show that the identification method proposed in this study may provide accurate tool wear condition identification based on machining process data, allowing for more flexible and precise cutting tool change decisions in machining.

Annotation table

ANN	Artificial Neural Network
SVM	Support Vector Machine
HMM	Hidden Markov Model
GLCM	Gray-level Co-occurrence Matrix
WVMD	Whitening Variational Mode Decomposition
PSO	Particle Swarm Optimization
ELM	Extreme Learning Machine
BPNN	Back Propagation Neural Network
PLSR	Partial Least Squares Regression
DL	Deep Learning
LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
IE-SBiGRU	Informer Encoder and Stacked Bidirectional Gated

Recurrent Un	
SBiGRU	Stacked Bidirectional Gated Recurrent Un
DKAE	Deep Kernel Autoencoder
PHM2010	American Society for Forecasting and Health Management
Attention-CNN	Single cutting force information and channel-spatial attention mechanism Convolutional Neural Network
Multi-CNN	Convolutional Neural Network founded on information fusion from many sources
Multi-Resnet	Residual Neural Network founded on information fusion from many sources
Multi-Attention-CNN	Convolutional Neural Network based on information fusion from several sources and a channel-spatial mechanism for attention
Multi-Attention-Resnet	Multi-source information fusion and channel-spatial attention force mechanism Convolutional Neural

* Corresponding author at: Key Laboratory of Advanced Manufacturing and Intelligent Technology, Ministry of Education, Harbin University of Science and Technology, Harbin 150080, China.

E-mail address: easy198766@126.com (R. Li).

<https://doi.org/10.1016/j.rcim.2024.102741>

Received 11 November 2023; Received in revised form 3 February 2024; Accepted 16 February 2024

Available online 24 February 2024

0736-5845/© 2024 Elsevier Ltd. All rights reserved.

Network

1. Introduction

The upkeep of machinery is crucial for production efficiency in the realm of intelligent manufacturing. Reduced production efficiency and financial losses are direct results of mechanical equipment failure. The cutting tool, known as the CNC machine tool's "tooth", is the direct execution end of mechanical cutting. CNC machine tools are a significant research object in equipment maintenance. The component of a CNC machine tool that is most susceptible to wear and damage is the tool. It has a great impact on the performance of the workpiece and the efficacy of the machine tool [1,2]. The condition of tool greatly influences the surface of the product's accuracy [1,2]. A machine tool may be harmed by severe cutting tool wear and even shut down accidentally. Additionally, the estimation of cutting tool state is typically cautious. The cutting tool is regularly replaced, which affects production efficiency and raises cutting tool costs. For data-driven intelligent manufacturing processes, tool wear condition monitoring technology is one of the most important factors affecting the intelligent development of modern machinery industry [3,4]. According to the research, the tool-caused downtime for machine tools accounts for 20 % of the total downtime [5]. The price of cutting tool and the price of renewing tool accounts for around 3–12 % of the total cost of processing output [6]. The growth of the mechanical manufacturing business is severely constrained by the production factor cost of mechanical cutting. To improve the core competitiveness of production, cutting tool wear condition monitoring technology is thus a crucial area for research and development [7].

More intelligent sensors and high-speed acquisition devices are being used in production systems today thanks to the development of new technologies like big data and artificial intelligence [8]. It is explored the link between the signal of the machining procedure and tool wear. The monitoring of wear state is achieved through deep data mining of machining process data [9]. This provides a suitable tool change strategy for the machining process.

Multi-source data generated during the cutting process were collected. Through big data analysis and other technologies, the data is deeply mined. A nonlinear mapping between cutting information and the condition of tool fatigue are determined. Due to the complexity and similarity of cutting process signals, it is not a simple task to identify changes related to cutting tool wear. To solve these challenges, a huge number of researchers employ machine learning-based methodologies. The commonly used machine learning models include ANN, SVM, HMM, Ensemble Learning Algorithm, etc. Bhat et al. [10] proposed the tool wear identification that uses the HMM, which classifies the tool wear state using features taken from the machining surface image's GLCM. With the goal to identify and filter out signals of sensitivity and efficiently prevent mode aliasing, Wei et al. [11] created a model for the breakdown of force signals using WVMD. The ideal feature subset is produced using the joint information entropy technique, and the tool wear state classification is accomplished using the ideal path forest algorithm. The wear state identification and prediction framework model using the least squares SVM, PSO, and superposition sparse denoising autoencoder was proposed by Xie et al. [12]. In terms of prediction accuracy, it outperforms ELM, BPNN, and PLSR. Sun et al. [13] adopted an improved support vector machine method, which used the effective feature set extracted from acoustic emission signals as input, learned each tool state information through two regularization parameters, and used five-fold cross validation to adjust the parameters. Sun et al. [14] identified effective signal features through the automatic correlation of Bayesian framework, used SVM to finish identifying the tool wear condition. With respect to achieve online learning and tool condition monitoring achieve online learning and tool condition monitoring, Liu et al. [15] presented an incremental learning algorithm-based ellipsoid network model, presented the Fast correlation filter method for adaptively selecting the

minimum redundant features, and built an incremental classifier. Wang et al. [16] constructed a tool state identification model based on relevance vector machine, which enhanced the wear state identification model's robustness.

In the area of identifying cutting tool wear state, machine learning models have shown considerable success. However, the state of wear identification network using machine learning usually has the shortcomings of shallow network structure, small scale and weak generalization ability. The depth mapping between the signal and the cutting tool wear state is difficult to identify. At the same time, its performance is heavily dependent on signal processing techniques and expertise. This may lead to underutilization of mechanical processing signals and difficulty in adapting to complex and variable processing scenarios.

The classification model based on DL has a strong ability to analyze complex nonlinear relationships [17]. It can mine the potential structural relationships between data by automatically learning the deep effective features of data [18], allowing the model to automatically discover and characterize feature information [19]. The generalization ability of DL is stronger than that of machine learning method, and the robustness is strong. It is very suitable for processing a large number of original data and can adapt to the complex and changeable processing environment [20]. DL techniques such as LSTM [21] and CNN [22] have been explored in the qualitative and quantitative monitoring of cutting tool wear, and can learn complex mapping relationships from original signals [23]. Ma et al. [24] proposed a deep connected Restricted Boltzmann machine using vibration and acoustic emission signal fusion for assessing cutting tool health state. Li et al. [25] utilized the IE-SBiGRU and it network as the backbone and the SBiGRU to improve the local connection of the characteristic order of a half cascade in order to forecast wear. Cao et al. [26] separated the original signal into information-specific frequency bands, used CNN to realize state classification. Sun et al. [27] proposed a technique using DL for predicting the condition of in-process tools. The method uses the original signal obtained in the machining process to construct the residual CNN, employs the correction according to median and the correction according to median to enhance accuracy so as to accomplish monitoring the condition of tools during the procedure of machining. Xu et al. [28] combined multiple machining process signals for feature fusion, and used DL model to identify cutting tool wear state. Ou et al. [29] proposed a DKAE feature learning method based on grey wolf optimizer optimization. By capturing spindle motor current signals from multiple channels along the X, Y, and Z axes for intelligent identification of wear state, the identification accuracy was greatly improved. Ou et al. [30] proposed an overlay denoising Autoencoder method using Online Sequential ELM to collect three-level current signals of CNC machine tool spindle in the process for intelligent identification of state. Li et al. [31] introduced a monitoring approach that utilizes data-driven techniques to identify and quantitatively anticipate cutting tool wear. This method is based on the fusion of radar chart features. A machine learning approach called Adaboost decision tree ensemble learning is developed to efficiently and correctly classify the present wear condition.

The cutting tool wear state identification based on DL method is mainly realized by machining process signals. However, Tool wear is time-varying information during machining, and the changing process of tool wear is nonlinear, the instantaneous change is fast, and it is difficult to predict. Redundancy and absence in the acquisition of tool wear information, and the recognizability of tool wear areas are the causes of misjudgment of tool wear, which is the result of a variety of factors [32,33]. Tool wear state identification is limited by the following shortcomings. (1) Mainstream DL models mainly use CNN and LSTM neural networks to classify cutting tool wear states. Most models use a single signal as the input that the model receives, but the single signal has certain limitations. This means that although they can capture the complex nonlinear relationship in the original data, they cannot achieve the complementarity of cutting tool wear information, and cannot achieve sequential reasoning [34]. (2) There is a fuzzy area between

various stage of cutting instrument wear. In the fuzzy area, it is challenging to distinguish the cutting instrument wear state model. If stage of severe wear is judged wrong, no relevant measures are taken. This will lead to a decrease in the quality of the workpiece, the scrap of the workpiece or even the shutdown of the equipment. (3) In DL, the more complex the neural network structure is, the more network layers and neurons need to be built. The more information the neural network can process. Complex models incur significant computational costs when dealing with large amounts of data. Given the aforementioned challenges, the mechanism of attention can process multi-dimensional information simultaneously, selective attention to important information, and suppression of useless information [35–38]. Data fusion from several sources mitigates the deficiencies associated with a solitary sensor with respect to data volume, diversity, and richness. A multi-source information fusion module is established by using multi-sensor signal as input. This improves the reliability and robustness of the system. Compared with single sensor, multi-sensor information fusion can realize the complementarity of tool wear information and state more accurately. By introducing mechanism of multi-dimensional attention into DL. A multi-dimensional and multi-channel attention mechanism is constructed, and it is reasonably applied to features. This can explore a wealth of hidden information. The network's information processing capacity has been enhanced. The accuracy of neural network identification is improved.

This research proposes an intelligent tool wear condition detection approach using machining process monitoring data, as shown in Fig. 1. Using continuous wavelet transform, the transformation of cutting signal and time-frequency image is completed. A single signal input cannot provide comprehensive identification information for the model. The reliability and accuracy of tool wear monitoring are improved by multi-source information fusion. This realizes the complementarity of processing information in time and space. The fuzzy regions of various wear stages are the primary source of tool wear condition misjudgment. Channel-spatial mechanism for attention is constructed. This makes it possible to selectively learn about the various related characteristics of cutting tool wear. The ambiguous area between varied phases of wear is reduced. The following are the main achievements of this research.

1. Through continuous wavelet transform, the transformation from the cutting process signal to the time-frequency image of the cutting process is realized. The time-frequency plot reflects the relationship

between frequency and time of the signal of the cutting process. In this paper, the middle region of the cutting signal is intercepted as the data training object. Convolutional neural networks are used as the basic architecture of the model. This can improve the accuracy of cutting tool wear identification.

2. The attention of channel-spatial module is introduced to learn the correlation between the channel and the space. Channel-spatial mechanism for attention is introduced into tool wear feature extraction. This enables adaptive adjustment of channel and spatial weights. Screening was done on the pertinent state condition characteristics. The precision of model classification and the stability of the model training process may both be enhanced by the channel-spatial attention method. The effect of ambiguous regions between varied phases of wear is diminished. The model still has high identification ability in severe cutting tool wear stage.

3. A multi-source information-based intelligent detection model of tool wear status is proposed. At the serious wear phase, the wear value fluctuates too rapidly. The distribution of information contained in different cutting signals is quite different. Only a single signal can not provide comprehensive identification information for the model. To this end, the fusion of data from several sources module is established. The features of the force signal and the vibration signal are fused. This realizes the complementation of feature information.

4. This paper realizes the training of the model through the PHM2010 data set. Five cutting tool wear identification models were adopted for validation. Five evaluation methods have been used. This enables a comprehensive evaluation of the model of categorization. The model's superiority is demonstrated by comparison with the existing categorization model.

2. Model's fundamental architecture

2.1. Preprocessing of multi-source information

The model's information for training in this study was gathered from the PHM2010 [39]. In the machining process, the cutting force signal offers useful information on cutting tool wear. This signal is generated only when the cutting tool makes contact with the workpiece. It is particularly sensitive to variations in the state condition and rises with wear. The force signal of cutting is widely recognized as a reliable

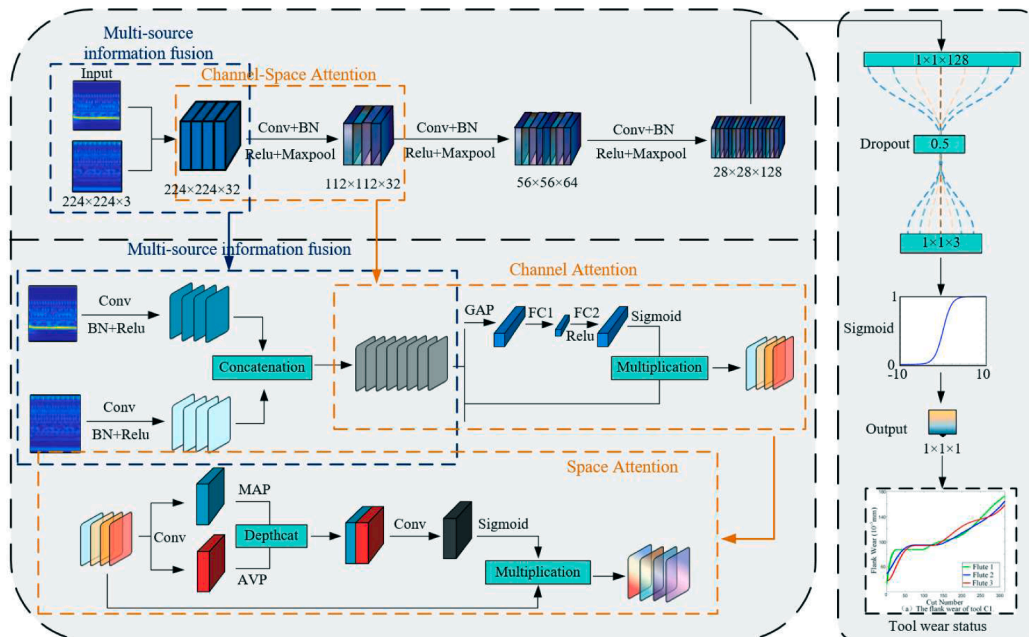


Fig. 1. Intelligent identification model for cutting tool wear status.

parameter for measuring and monitoring cutting tool wear [40]. The vibration signal reflects the machine tool vibration generated via means of direct touch between the tool and the workpiece during CNC machining. It shows a strong sensitivity to the cutting tool conditions [41]. A single sensor cannot capture all relevant information about tool wear. Force signals and vibration signals of cutting are utilized to identify states. The middle region of the processed signal is used as the training dataset. The influence of the cutting process and the cutter in and out process is reduced. The chronological properties of the machining signal do not provide an accurate depiction of the device wear states. Therefore, it is necessary to convert the machining monitoring signal into the corresponding time-frequency image for analysis. The preprocessing of the processing signal is shown in Fig. 2. This enables non-off-line monitoring of the tool condition. Continuous wavelet transforms time-domain signals into time-frequency pictures, which can capture the specific dependent upon time data characteristics locally [42]. An exact instantaneous change of the tool signal during cutting are provided by the processed time-frequency characteristics [43]. As a result, the model for recognizing states becomes more accurate.

2.2. Fusion of multi-source information

Multi-sensor signals have further details on wear of tool than single signals, which can enhance model identify precision. Because the time-frequency image of the original signal is utilized as the method's data for training. CNN is efficient and flexible for image processing. Therefore, the CNN parallel structure is adopted. The time-frequency picture characteristics of the force signal and the vibration signal are extracted and merged. This may greatly improve the accuracy of tool wear state identification. The CNN parallel structure part uses 2D CNN, whose

structure is shown in Fig. 3. The time-frequency picture of the force signal and the vibration signal are used as input. The convolution kernel slides over the time-frequency picture of the processed signal separately. Different feature maps are obtained by multiple convolution kernels to extract cutting tool wear state features. Parallel CNN can extract features from multiple different signals, concatenate different features, and realize the fusion of multiple information.

2.3. Feature extraction based on channel-spatial mechanism for attention

The key elements of tool wear are contained in the information time-frequency image after continuous wavelet treatment. However, the connection between tool wear characteristics is either strong or weak. The capability of the model to effectively detect the cutting tool wear condition is based on the model's high correlation features of wear. Although the weakly linked feature of tool wear helps to identify the wear state, it contributes less of training and increases the training variables. Consequently, the states identification model may learn the related aspects of tool wear in a targeted way by building a channel-spatial mechanism for attention [44,45]. The ambiguous areas of tool wear identification at different stages are reduced. The accuracy of identification is improved. The fundamental idea behind the attention mechanism is derived from the extracted features of wear. Tool wear feature values are evaluated separately in the channel and spatial dimensions, and important features receive greater weights. This improves the accuracy of state identification [46]. To train and update the model, the tool wear state identification model makes use of the high connection properties of tool wear. This enables tool wear state identification.

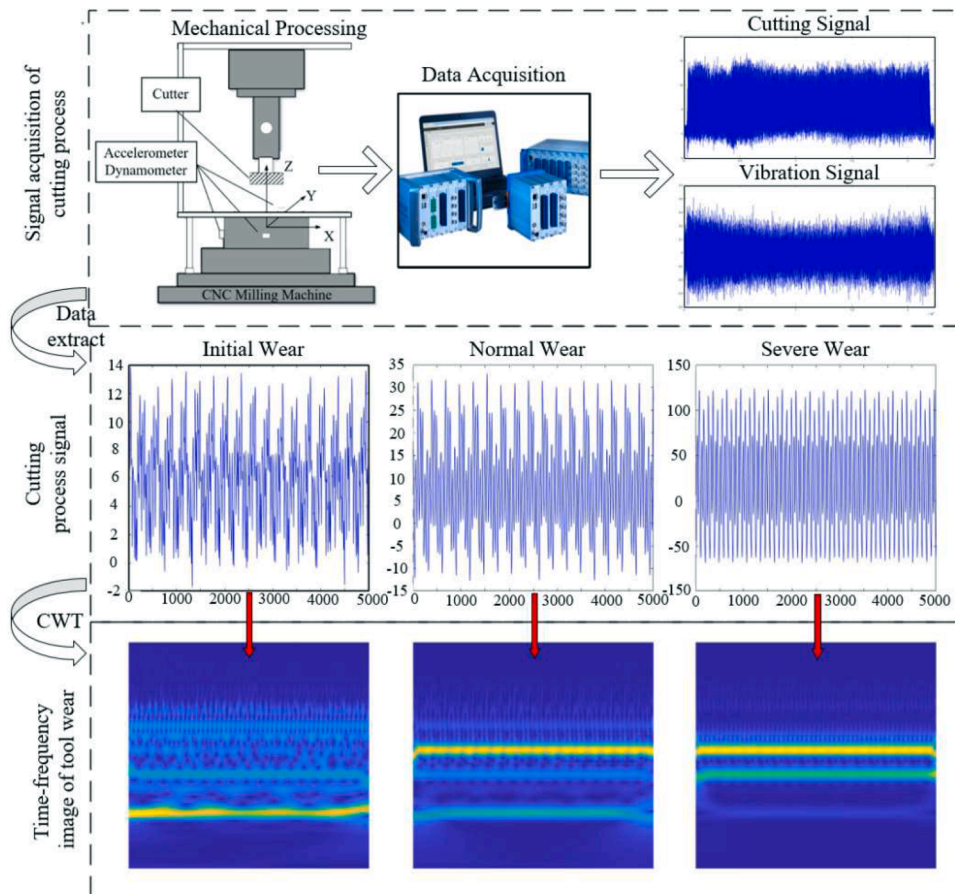


Fig. 2. Data preprocessing in cutting process.

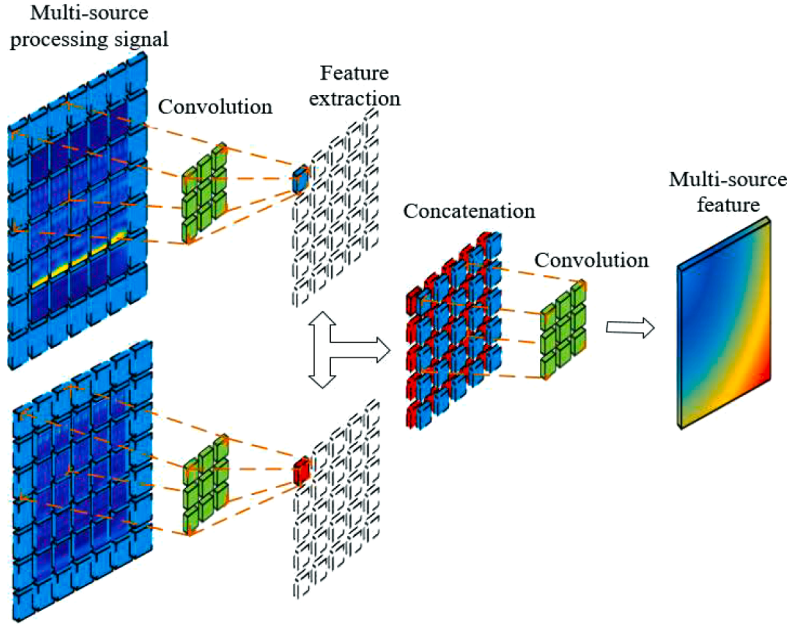


Fig. 3. Fusion of multi-source signals.

2.3.1. Weight adjustment of features in channel dimension

By employing the wear sensitive characteristics, the identification model determines the wear states. However, the redundant information in the model will increase the difficulty of model training, resulting in increased model training time. Convolution is used to generate the multi-channel wear feature map. As the amount of training data rises, the weights of the multiple channels will be automatically modified by the input of the time-frequency. The precision of tool wear state identification will be impacted by this. Different channels' cutting tool wear-sensitive characteristics have varying degrees of correlation impacts on tool state identification [38,47]. Some cutting tool wear characteristics are critical in determining wear. As observed in Fig. 4, the channel attention process is separated into three parts.

The CNN only works in a section of the area when extracting the sensitive features. This leads to difficult connections between different channels and smaller receptive fields between different channels. The feature channel is compressed by the squeeze operation. The tool sensitive feature is compressed into a global feature.

$$X^{H \times W \times M}_{train} = \{X_1, X_2, \dots, X_M\} \quad (1)$$

$$F_{sq}^i = GAP(X_{train}^i) \quad (2)$$

$$F^{1 \times 1 \times M}_{sq} = \sum_{i=1}^M F_{sq}^i \quad (3)$$

Where $X^{H \times W \times M}_{train}$ is the sensitive feature after convolution, with M feature channels and a feature map size of $H \times W$. F_{sq}^i is the global

feature by global average pooling of the i th feature map. GAP calculates the average of each feature map in the neural network. $F^{1 \times 1 \times M}_{sq}$ corresponds to M channels on a global feature scale.

The dimensionality of the feature map is reduced along the dimension of channel by the process of excitation operation in the fully connected layer FC_1 . The Relu activation function realizes the nonlinearity. The operation FC_2 serves to restore the channel size of features in its original size. The function of sigmoid is utilized to transform the processed vector of features into a weight vector that is normalized between the range of 0 and 1, with the value $S^{1 \times 1 \times M}$. The weight values for several sensitive traits are generated. The use of FC_1 and FC_2 possesses superior nonlinear fitting capability, which can suit the intricate relationship between the channels more precisely. The expression in question is as follows.

$$FC_1 = W_1 \times F^{1 \times 1 \times M}_{sq} \quad (4)$$

$$FC_2 = W_2 \times Relu(FC_1) \quad (5)$$

$$S^{1 \times 1 \times M} = Sigmoid(FC_2) \quad (6)$$

This is shown in Fig. 3, $W_1, W_2 \in R^{M \times M}$, M represents the number of tool sensitive features. r represents the dimension reduction factor. The feature vector $S^{1 \times 1 \times M}$ represents the sensitive characteristics for M channels. The activation function utilized is the sigmoid function. The expression in question is as follows.

$$S(x) = \frac{1}{1 + e^{-x}} \quad (7)$$

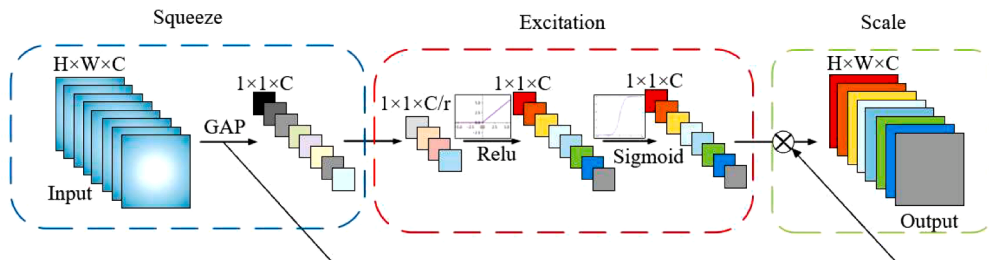


Fig. 4. Channel attention mechanism.

Finally, through the scale operation, the pertinent expressions are presented below. The channel weight $S^1 \times 1 \times M$ is combined with the sensitive feature $X^{H \times W \times M}_{train}$ from the original channel to produce a new sensitive feature $\hat{X}^{H \times W \times M}_{train}$. This integration enhances the model's ability to distinguish between the sensitive features of various channels.

$$\hat{X}^{H \times W \times M}_{train} = X^{H \times W \times M}_{train} \times S^{1 \times 1 \times M} \quad (8)$$

2.3.2. Weight adjustment of features in the spatial dimension

The integration of a spatial attention mechanism enhances the identification model's capacity for spatial discrimination. This enables focused attention on strongly information in various regions of the characteristics [34], as seen in Fig. 5. First, The dimension of the channel is compressed, and the characteristics of numerous channels are combined using a kernel with a size of [1,1].

$$X^{H \times W \times 1}_{train} = f^{1 \times 1} * (X^{H \times W \times M}_{train}) \quad (9)$$

Where $f^{1 \times 1}$ is the convolution core; $*$ denotes operation of convolution; $X^{H \times W \times M}_{train}$ is feature map; $X^{H \times W \times 1}_{train}$ is the integrated feature of a channel formed by a correlation of M channels, with size $H \times W$.

On the feature maps, max pooling and avg pooling of the channel's dimension are conducted to generate the feature matrix F_{max} after max pooling and the feature matrix F_{avg} after average pooling. Combining F_{avg} and F_{max} together gives the new feature Figure. $F^{H \times W \times 2}_{train}$. Finally, the new feature $F^{H \times W \times 1}_{train}$ is obtained by operation of convolution. The feature matrix undergoes a transformation using the Sigmoid function, resulting in a normalized coefficient ranging from 0 to 1. This normalized coefficient, denoted as M_s , is the weight coefficient associated with the spatial attention mechanism. By multiplying weight coefficient with the input feature $X^{H \times W \times M}_{train}$, the sensitive feature $\hat{X}^{H \times W \times M}_{train}$ corrected by spatial attention can be obtained. The expression in question is as follows.

$$F_{avg} = Avgpool(X^{H \times W \times 1}_{train}) \quad (10)$$

$$F_{max} = Maxpool(X^{H \times W \times 1}_{train}) \quad (11)$$

$$F^{H \times W \times 2}_{train} = F_{avg} \oplus F_{max} \quad (12)$$

$$F^{H \times W \times 1}_{train} = f^{1 \times 1} * (F^{H \times W \times 2}_{train}) \quad (13)$$

$$M_s = Sigmoid(F^{H \times W \times 1}_{train}) \quad (14)$$

$$\hat{X}^{H \times W \times M}_{train} = X^{H \times W \times M}_{train} \times M_s \quad (15)$$

The feature map's max pooling and avg pooling are represented by the terms Maxpool and Avgpool. F_{avg} and F_{max} are avg pooling and max pooling of wear features. After multi-channel fusion, $X^{H \times W \times 1}_{train}$ is the one-channel functionality; $\oplus$ denotes the process of combining two feature matrices; $F^{H \times W \times 2}_{train}$ is the two-channel characteristic in parallel;

M_s is the normalized coefficient of spatial attention feature; $\hat{X}^{H \times W \times M}_{train}$ shows wear features corrected by the attention mechanism.

2.3.3. Cutting tool wear feature extraction

The cutting tool wear condition detection model's pooling layer and convolutional layer are joined by the channel-spatial mechanism for attention. The tool wear condition identification model is endowed with a selective learning capacity and is able to extract highly correlated characteristics that are associated with the wear state. As the quantity of layers of convolution is augmented, the feature picture of the third layer becomes progressively smaller during feature extraction using the tool state categorization model. When incorporating the channel-spatial mechanism for attention into this layer, it is observed that the spatial attention tends to concentrate on a very tiny feature region, which may not effectively capture the crucial information pertaining to cutting tool wear characteristics. The inclusion of the channel-spatial mechanism for attention in all convolutional layers result in an excessive number of parameters and prolonged training duration, hence potentially compromising the model's anti-interference capability. The attention mechanism is only utilized to load the network's first layer, which performs best, and isn't added to each convolutional layer. Literature [48] has carried out research on attention mechanism based on machine vision, and analyzed the emphasis and inhibition of feature learning achieved by attention mechanism, confirm that can focus on the target image correctly.

Through the channel-spatial attention method, the model exploits the fusion of multi-source information in the cutting process to identify the deep aspects of tool state. Fig. 6 shows the final channel-spatial mechanism for attention. After the dimensionality reduction and fusion of the multi-source signals F_{filter} in the cutting process through the convolution kernel, the feature $X^{H \times W \times M}_{train}$ is obtained. Channel and spatial mechanism for attention are used to analyze the characteristics of wear stages. The model can identify strong correlation features in wear state. In the subsequent training, the model is updated according to the strong characteristics with higher value. The expression in question is as follows.

$$X_{ca} = X^{H \times W \times M}_{train} \times s \quad (16)$$

$$X_{sa} = X^{H \times W \times M}_{train} \times M_s \quad (17)$$

$$\tilde{X}_{train} = X^{H \times W \times M}_{train} \times s \times M_s \quad (18)$$

Where, X_{ca} is channel dimensional feature; X_{sa} is spatial dimensional feature; s is the correction coefficient of channel attention mechanism; M_s is the correction coefficient of spatial attention mechanism; $\tilde{X}_{train}$ shows the corrected cutting tool wear feature after sequential combination of channel attention and spatial attention.

2.4. Convolutional neural networks

The cutting tool wear state model's fundamental architecture

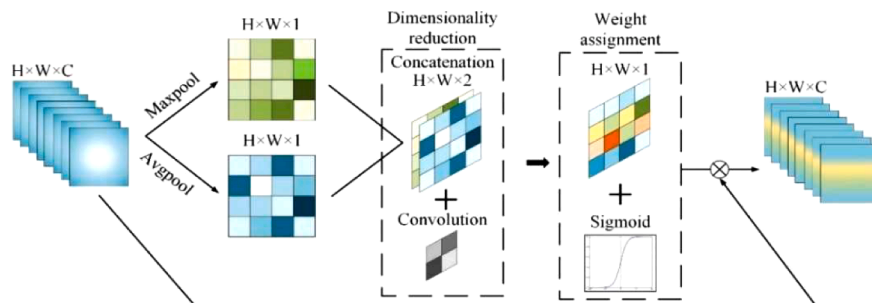


Fig. 5. Spatial attention mechanism.

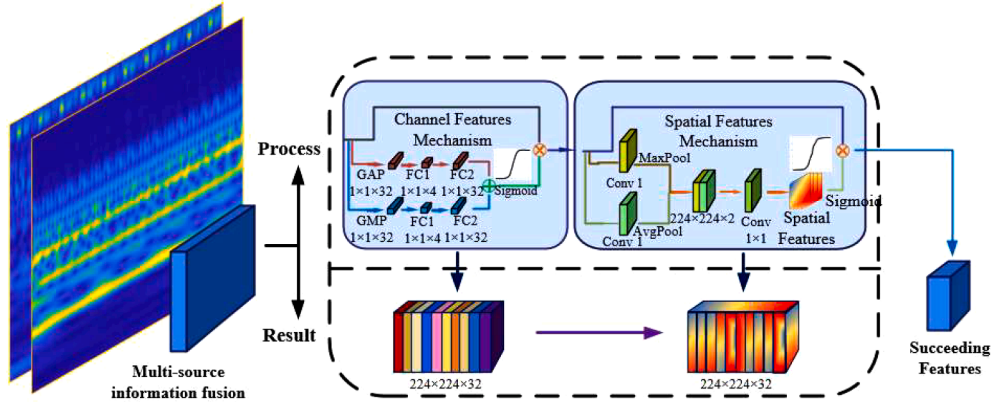


Fig. 6. Channel-spatial multiple attention mechanism.

comprises two fully linked layers and three layers of convolution. The tool wear state identification is realized by adding feature fusion and channel-spatial mechanism for attention modules to the basic framework. The normalization layer reduces the time to train the network by normalizing the input of the neural network to 0 mean and unit variance. The activation layer Relu realizes the nonlinearity of the cutting tool wear state information. The layer of pooling is decreasing the dimensionality of individual feature maps, while simultaneously preserving crucial information and mitigating the risk of overfitting in the model. Fully connected to contain 128,3 neurons respectively. The data of the fully connected layer are used to identify wear state through the Softmax categorization function.

The neural network's input consists of a time-frequency picture with dimensions of $224 \times 224 \times 3$, where the values 224×224 represent the actual dimension of the image, 3 means that the input time-frequency image is an RGB image. There are three layers of convolution in the tool wear state identification model. Convolution kernel size is all [3,3], which ensures the extraction of tool sensitive features and the training time, and improves the receptive field. The expression in question is as follows.

$$F_{ilter} = \{x_1, x_2, \dots, x_c\} \quad (19)$$

$$X_i = f \left(\sum_{k=1}^C W_k^i \times F_{ilter}^i + B_i \right), i \in [1, C] \quad (20)$$

Where X_i is the feature matrix that comes out of the i th kernel; W_k^i is the k th channel's i th kernel; F_{ilter}^i displays the i th channel's input cutting tool wear feature map; B_i is the bias value of the i th kernel; The activation function is f , which may be stated as follows.

$$ReLUx = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases} \quad (21)$$

The worn features of the cutting tool that have been extracted are then compressed using a pooling layer. This layer decreases the dimension of these characteristics while keeping the information contained inside them. It additionally lowers the characteristic of output's sensitivity to distortion. For the purpose of a relationship between wear characteristics and states, two completely linked layers were built to handle the 128 retrieved wear features. The input to this layer consists of the features that are sensitive to the cutting tool. Each neuron in this layer is linked to all neurons in the preceding layer, allowing for the integration and extraction of associations between the aspects linked to wear states.

2.5. Model evaluation metrics

The classification model's diverse parameters are acquired via the

process of training the model. A network model of higher complexity necessitates a larger amount of data for the purpose of training. With the goal to evaluate the discrepancy in performance among various kinds, five performance measures are employed to evaluate the models' performance. These are Confusion Matrix, Accuracy, Precision, Recall, and F1 score.

$$Accuracy = \frac{True\ Positive + True\ Negative}{True\ Positive + True\ Negative + False\ Positive + False\ Negative} \quad (22)$$

$$Precision = \frac{True\ Positive}{True\ Positive + False\ Positive} \quad (23)$$

$$Recall = \frac{True\ Positive}{True\ Positive + False\ Negative} \quad (24)$$

$$F1\ score = \frac{2 \times Recall \times Precision}{Recall + Precision} \quad (25)$$

The Accuracy metric provides an assessment of the categorization model's overall performance. Precision represents the accuracy of identification at various levels of wear, which means the proportion of correct model analysis in the predicted value. Recall represents the identification of various stages, which is the proportion of data correctly deemed as positive out of all information having true positive value. F1 score is the harmonic mean of precision and recall, with greater values indicating superior performance in categorization.

3. Experimental data and training parameter setting

3.1. PHM2010 data set

The data from experiments employed in this study for tool wear condition categorization model is the PHM2010 dataset. This is shown in Fig. 7. The milling cutting tool is a 3-tooth ball end carbide milling cutter. The transverse cutting depth was 0.125 mm. The longitudinal cutting depth was 0.2 mm. The feed of the knife is 0.001 mm. The process of milling was conducted using a high-speed Computer Numerical Control (CNC) equipment (Rodgers Tech RFM760). The high-speed CNC machine tool's speed of the spindle is 10400 r/min. The feed rate is 1555 mm/min. The stainless steel (HRC52) workpiece was subjected to milling using a 6 mm three-slot tungsten carbide ball head. This milling process was conducted under dry circumstances, and the workpiece was milled row by row. The milling process was repeated for six complete life cycle tests. The Kistler three-term force meter is installed between the workpiece and the processing table to collect the milling force signals in the y, x, and z directions. The material used for end milling is square in shape, with each milling having a length of 108 mm. On the workpiece, Kistler piezoelectric accelerometers capture

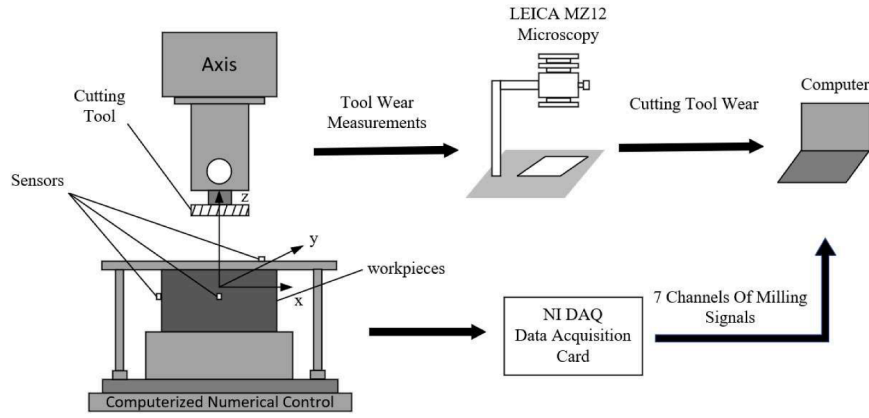


Fig. 7. Experimental setup.

vibration information in the y, x, and z axes. The collected 6-channel signals were processed through a Kistler multi-channel charge amplifier. Collecting information at a sample rate of 50 kHz is achieved by means of NI DAQ data acquisition card. The acquisition data consists of the cutting force signal, vibration signal, and acoustic emission signal along the x, y, and z axes. The tool wear after each cutting cycle was recorded in units of 10^{-3} mm. After each knife movement, the wear width (VB) in every groove was measured using the LEICA MZ12 microscope. Among the six groups of test data collected, the C1, C4, and C6 data contain the wear amount.

3.2. Analyze the PHM data set

The categorization of wear status following EM algorithm classification is displayed in Table 1. When training the model, each stage's data is separated into the training set and test set in a 7:3 ratio. This is done to guarantee that the model receives enough data during the training phase, but also to avoid the training data appear in the test set, causing data pollution. The possibility for overfitting of the identification model is reduced.

Since only C1, C4 and C6 contained the label of cutting tool wear in the data set, the model in this paper was tested based on three groups of data. Each trial contained data from 315 knife moves. As shown in Fig. 8 (a), each cutting tool wear label contains the wear values of three grooves. The wear amount of each wear groove is different and nonlinear. It is not accurate to judge the cutting tool wear by only one groove. When establishing tool wear criteria, the average wear value of all segments involved in the cutting operation is often utilized. The average cutting tool wear for the three grooves of the C1, C4, C6 datasets is depicted in Fig. 8(b). As seen in Fig. 8(c), the phenomenon of cutting tool wear may be categorized into three distinct stages, initial, normal, and severe wear. In order to reduce the influence of the tool wear data set on the accuracy of the model, the wear state is classified through the utilization of the expectation maximization algorithm (EM). Fig. 8(d) depicts the end outcome. The clustering method receives the tool VB as input and outputs the wear state.

3.3. Training parameter settings

The model in this research was trained using the software

Table 1
Partition of C1, C4, and C6 datasets.

Dataset	Initial wear	Normal wear	Severe wear
C1	1–47	48–146	147–315
C4	1–135	136–204	205–315
C6	1–81	82–188	189–315

Matlab2021a and a graphics card configuration of RTX3090. To validate the proposed model's performance gains, the same training parameters and data set settings are used for each model. To assess cutting tool wear state, the subjects are trained on the same computer platform: Multi-CNN, Multi-Resnet, Multi-Attention-CNN, Multi-Attention-Resnet, Attention-CNN. Among them, the inherent structure of CNN can well realize the extraction of two-dimensional wear features. Resnet adds residual blocks to enhance the expression ability of deep neural networks. Both have three convolutional layers and two fully linked layers as its fundamental structure. The relevant network and training parameter Settings are the same for each model. Table 2 displays the basic structural parameters.

During the training period, the setting of learning rate, training batch and iteration times has various effects on the accuracy of the model. The high initial learning rate will make it difficult to achieve the global optimum in the process of training the model. Small learning rate will make the training model take a lot of time. Larger training batches can make the training process less volatile. It can improve the stability of training and save training time. However, mini-batch training will bring more fluctuations in training process, and the randomness of weight updates is stronger, which improves the generalization ability of the model. An inadequate number of iterations can result in the model being underfitted, whereas an excessive number can cause the model to be overfitted, which will reduce the accuracy of the model. The training procedure is depicted in Fig. 9 under various training conditions.

Different training conditions will lead to different identification accuracy of the model. The observation from the training process indicates that as the amount of training iterations increases, the model's identification accuracy eventually stabilizes. When using the serial number 1 scheme for model training, the model's accuracy exhibits a progressive fluctuation as the duration of training increases. This is because too many iterations lead to overfitting of the model. Schemes 2 and 3 are used for training. When the size of the batch is 256, the model's training is more stable, but its generalization capacity is reduced. By adjusting the initial learning rate of the training process and using scheme 4 for training, the performance of the model is lower than that of scheme 2 and 3. This is due to the high initial learning rate, which makes the model unable to establish the mapping between machining process data and wear state. Based on the above analysis, in order to enable the model to obtain higher generalization ability with less training time. Accuracy of the model to extract cutting tool wear characteristics is improved. Scheme 5 is used as the training strategy of the model.

- (1) A rate of dynamic learning was implemented. A consequence of the constant rate of learning is that the model will converge to the local optimal state. After reaching the corresponding number of iterations, the deep information of cutting tool wear cannot be

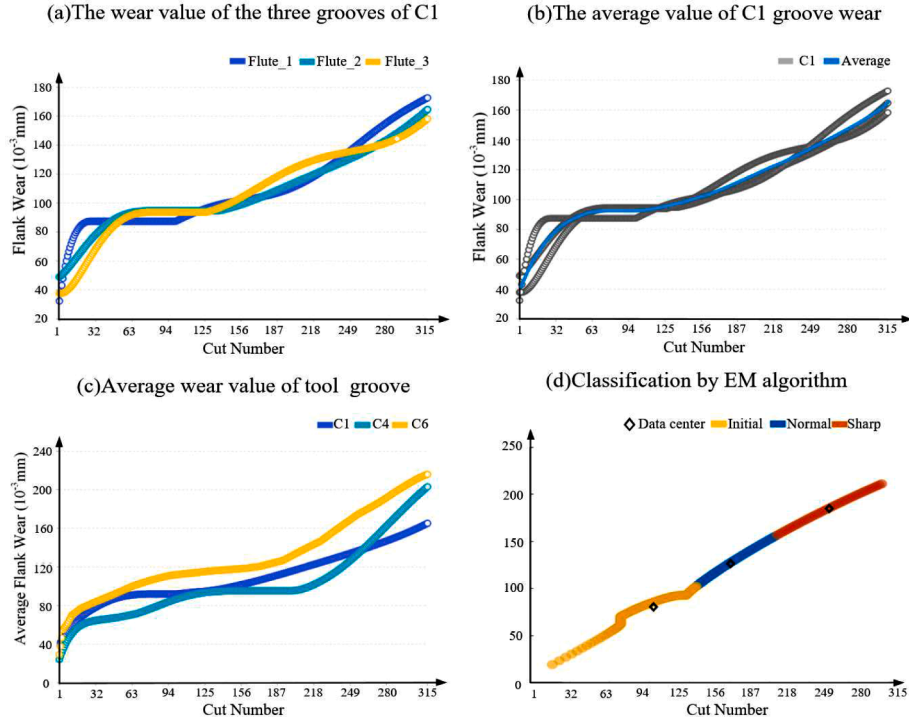


Fig. 8. Analysis of cutting tool wear curve.

Table 2

Basic parameters of the neural network model.

Model	Layer number	Description	Parameter Settings
Basic structure of neural network	Layer1	2D-Convolution	filters = 16, kernel size = 3, strides = 1, padding = same, output=224 × 224 × 16, activation = ReLu
	Layer2	Concatenation	output=224 × 224 × 32
	Layer3	Max Pooling	pool size = 3, strides = 2, padding = same, output=112 × 112 × 32
	Layer4	2D-Convolution	filters = 64, kernel size = 3, strides = 1, padding = same, output=112 × 112 × 64, activation = ReLu
	Layer5	Max Pooling	pool size = 3, strides = 2, padding = same, output=56 × 56 × 64
	Layer6	2D-Convolution	filters = 128, kernel size = 3, strides = 1, padding = same, output=56 × 56 × 128, activation = ReLu
	Layer7	Max Pooling	pool size = 3, strides = 2, padding = same, output=28 × 28 × 128
	Layer8	Fully Connected	output=1 × 1 × 128
	Layer9	Dropout	dropout=0.5
	Layer10	Fully Connected	output=1 × 1 × 3
	Layer11	SoftMax	

further mined. Hence, after 30 iterations, the learning rate is configured to decrease to 1/10 of its initial value.

- (2) The amount of iterations for the training process is determined as 100 in order to mitigate the risks of underfitting and overfitting in the model.

- (3) The Adams optimizer is used to train the model. The Adams optimizer's adaptive optimization capability may adaptably change the model network weights and works well in the DL model.
- (4) L2 regularization is employed to mitigate the issue of over-fitting in the model, resulting in a smoother trajectory of the training process. The function of loss utilized in the model is a cross-entropy, which can be expressed as follows.

4. Analysis of experimental outcomes and model performance

The primary objective of this research is to assess the efficacy of the tool for cutting wear detection model through five evaluation indicators such as Confusion Matrix and Accuracy. The Confusion Matrix of the model is presented in Fig. 10. Because there are many data sets in the state of severe wear and tear, and the data in the severe stage fluctuates greatly, there are some abnormal points. The data between different wear stages contain the cutting tool wear feature information of adjacent wear stages, it is easy to lead to misjudgment of the model. Consequently, there arises a necessity for the model to effectively discern the data pertaining to the stages of severe wear and regular wear. The Confusion Matrix provides the classification outcomes of the model for each state of wear. In most cases, the tool should be replaced before the severe wear occurs, which requires the model to have a higher identification rate for the cutting tool under severe wear. In this paper, the model Multi-Attention-CNN has four sample identification errors in 283 samples of the training set. The identification accuracy was 98.6 %. All of the samples at the stage of severe wear have been accurately recognized.

For each model in the article, 12 model trainings are performed to remove the most and the worst results. This is done in an effort to decrease error, more effectively demonstrate the model's performance, and enhance persuasion. Fig. 11 shows the average results and their visualization after 10 trainings.

Table 3 shows the performance metrics after 10 iterations of model training. The Accuracy, Precision, Recall performance indicators of the Multi-Attention-CNN model proposed in this study are 0.982, 0.98, 0.98 and 0.979. The model under consideration demonstrates superior

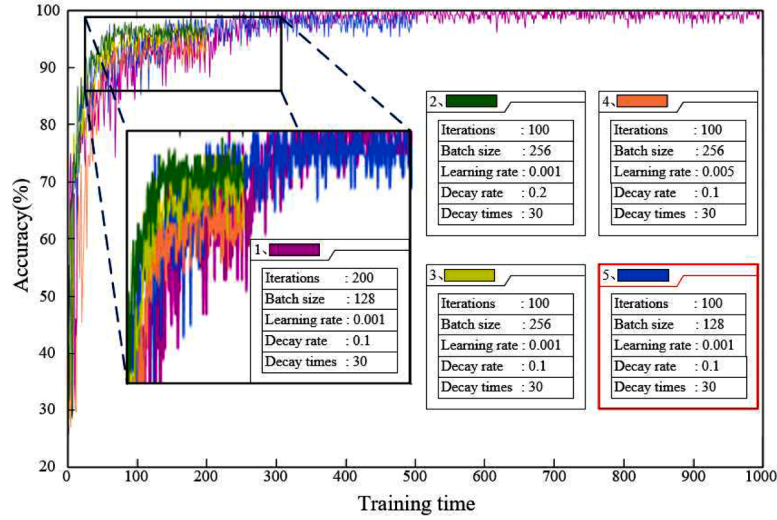


Fig. 9. The procedure by which models are trained under various conditions.

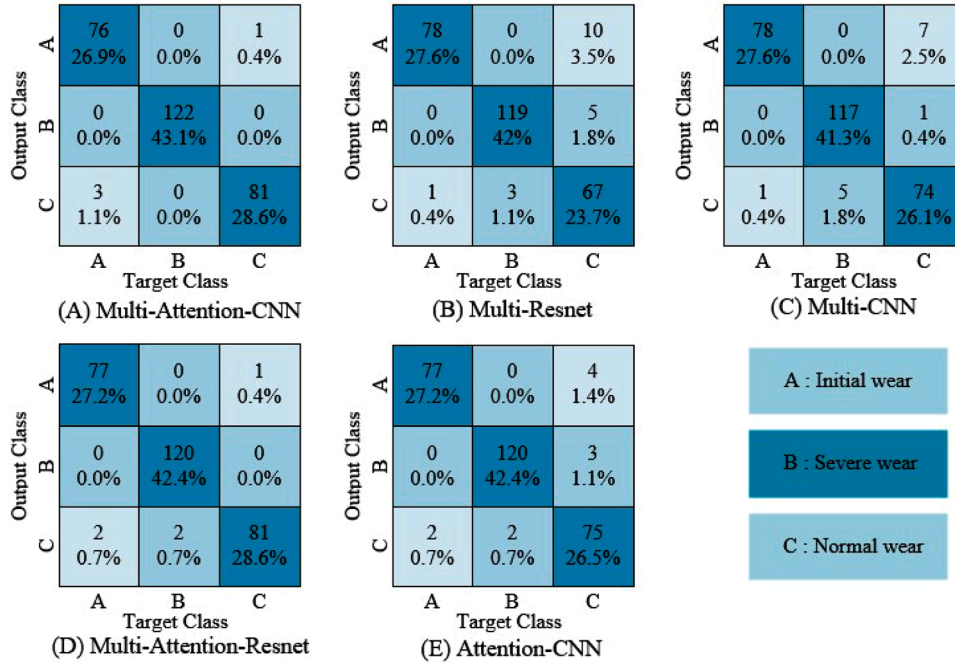


Fig. 10. Confusion Matrix.

performance compared to the other models.

According to the data in Table 3, the accuracy of the Multi-Attention-CNN method is 1.74 % and 2.15 % greater than that of the Attention-CNN method and the Multi-CNN method, respectively. However, in the recognition scenario of state, the recognition entering a severe wear state is particularly critical. The cutting tool with severe wear state is more harmful to be misjudged. And the normal wear state has a cross area with the initial and severe wear, misjudgment of cutting tool condition often occurs in this area. Hence, it is not justifiable to just evaluate the model's effectiveness based on the overall identification rate. In this paper, Precision and Recall need to be used for stage evaluation of wear status.

Fig. 12(a) depicts the initial wear stage, the Accuracy and Recall rate of the Multi-Attention-CNN model are 0.98, 0.975. Compared with the Multi-CNN model, the Accuracy and Recall rate are improved by 2.4 % and 2.9 %. The Attention-CNN's classification performance in early

wear is essentially similar with the model developed in this research. This indicates that multi-source information fusion contributes less in this wear stage, and Fig. 12(b) also confirms this rule. At this stage, the channel-spatial mechanism for attention gives useful information to the model in determining the tool state, which increases the identification efficiency and accuracy of the initial wear.

As seen in Fig. 13(a), the Accuracy and Recall rate of the Multi-Attention-CNN model in normal wear stage are 0.965, 0.972. Compared with the Attention-CNN model, it increased by 2.2 %, 2.2 %. The Accuracy and Recall rate of the Attention-CNN model are 1.7 % and 0.5 % higher than those of the Multi-CNN model. This shows that when model performs state identification in the normal wear stage, the contribution of the channel-spatial mechanism for attention is greater than that of information fusion from several sources. The combination of the two can substantially enhance the identification efficiency of the model during normal wear. Fig. 13(b) also shows that multi-source

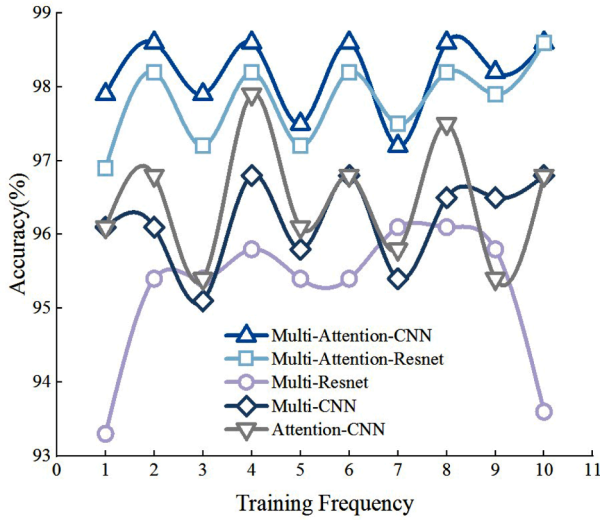


Fig. 11. Accuracy of Model 10 Trainings.

Table 3
Performance of 10 Model Trainings.

Model	Wear States	Evaluating Indicator			Accuracy
		Precision	Recall	F1score	
Attention-CNN	Initial	0.977	0.979	0.980	0.965
	Normal	0.943	0.950	0.946	
	Severe	0.980	0.967	0.973	
	Average	0.967	0.965	0.966	
Multi-CNN	Initial	0.956	0.946	0.951	0.962
	Normal	0.926	0.945	0.935	
	Severe	0.992	0.984	0.988	
	Average	0.958	0.958	0.958	
Multi-Resnet	Initial	0.942	0.946	0.944	0.952
	Normal	0.92	0.917	0.918	
	Severe	0.983	0.98	0.981	
	Average	0.948	0.948	0.948	
Multi-Attention-Resnet	Initial	0.979	0.968	0.973	0.978
	Normal	0.955	0.971	0.963	
	Severe	0.994	0.989	0.991	
	Average	0.976	0.976	0.976	
Multi-Attention-CNN	Initial	0.98	0.975	0.977	0.982
	Normal	0.965	0.972	0.968	
	Severe	0.994	0.993	0.993	
	Average	0.98	0.98	0.979	

information fusion is of limited help at this stage. Using multi-source information fusion and channel-spatial mechanism for attention alone cannot maximize the identification performance of the model.

The initial and normal stages of the tool wear is analyzed. The channel-spatial mechanism for attention has a greater improvement on the model. This is due to the fact that the cutting tool wear value fluctuates slowly in initial and normal stages of tool wear. At the same sampling frequency, the pheromone difference of tool wear characteristic information of cutting force and vibration is small. When the identification model is classified, the misjudgment of the tool state is mainly distributed in the intersection area of initial wear and normal wear, as shown in Fig. 14. Fig. 14(b) is the misclassification distribution of the Multi-CNN model in the initial and normal stages of cutting tool wear. At this wear state, the effect of fusion information from multiple sources is poor, the complementarity of information in time and spatial cannot be realized. Fig. 14(a) is a Multi-Attention-CNN model with channel-spatial mechanism for attention. The fuzzy regions between

distinct stages can be reduced using the channel-spatial mechanism for attention. Ensuring accurate detection of the tool's stage.

The cutting tool's identity in a severe wear state is particularly critical. The model evaluation parameters for this stage are shown in Fig. 15(a). The Accuracy and Recall rate of the Multi-Attention-CNN model are 0.994, 0.993. In the severe wear stage, the values for Precision and Recall of the Multi-CNN model are 0.992 and 0.984, respectively. The Accuracy and Recall rate of the Attention-CNN model were 0.98 and 0.967. It can be seen that the main contribution of the identification of severe wear phase comes from the information from several sources combined. Fig. 15(b) also proves this view. Through information fusion from several sources, the successful recall ratio of the model in the severe wear phase is improved. This reduces the absence of tool state identification during severe wear phases. The Recall of the heavy wear phase is increased. In the severe wear phase, although the channel spatial attention mechanism may boost the model's performance, the contribution of multi-source information fusion is greater.

The cutting tool's severe wear stage is investigated. At this stage, the approach is mostly based on information fusion from several sources, which may significantly increase the model's identification performance. This phenomenon occurs due to the significant alteration in the cutting tool wear value when it reaches the stage of severe wear. The energy span of collected detecting signal is high. At the same sampling frequency, the pheromone of tool wear characteristic information of cutting force and vibration is quite different. Different information provides complementary information in time and spatial for cutting tool wear characteristics through fusion. Therefore, it provides a guarantee for the identification of state in the severe wear.

The whole process identification of wear state is analyzed. Table 3 shows that the identification accuracy of the Multi-CNN and Attention-CNN models, respectively, is 0.962 and 0.965. Although the comprehensive performance of the two is the same, the Multi-CNN model has a high identification rate in the severe wear stage. The Attention-CNN model's accuracy in identifying the tool wear condition at all stages has been increased. The attention mechanism can reduce the tool wear cross area and enhance the global performance of tool state identification model. The comprehensive effect of adding attention mechanism to the basic network is better than that of adding multi-source information fusion. This further confirms that the main source of misjudgment in state identification of tool wear is the intersection area of different intervals of tool wear. The distribution of misjudgments in the 10 identification processes of the validation set in Fig. 16.

In summary, In the end, the intelligent cutting tool status identification model that is multi-source information driven presented in this article has higher identification performance. The Multi-Attention-CNN model presented in this paper has superior performance compared to previous models. As shown in Fig. 17, the Accuracy, Precision, and Recall rates are 0.982, 0.98, and 0.98. And it still has a high identification rate in each stage.

The channel-spatial mechanism for attention facilitates the allocation of weights based on the significance of tool wear characteristics. The model obtains adaptive selection ability. The cross area of different wear intervals is reduced, the model's overall performance is enhanced. Multi-source information fusion is adopted. In the severe wear stage, the model realizes the discrimination of tool state according to distinct information in the same stage, and improves the proportion of recall in the severe wear state.

5. Conclusion

In machining, the causes of misjudgment caused by cutting tool wear stage are different. The cutting tool wear value changes slowly during the initial wear and normal wear. The overlap between various phases of cutting tool wear is substantial, hence serving as the primary cause for the misinterpretation of tool condition. In severe wear stage, the wear value changes too fast, the information distribution contained in

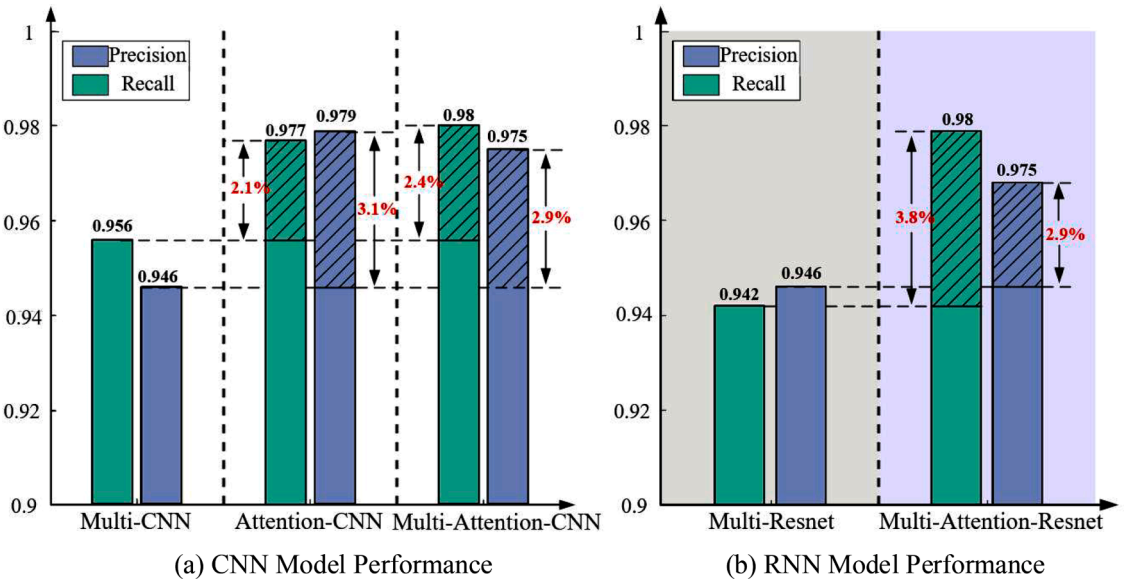


Fig. 12. Initial wear.

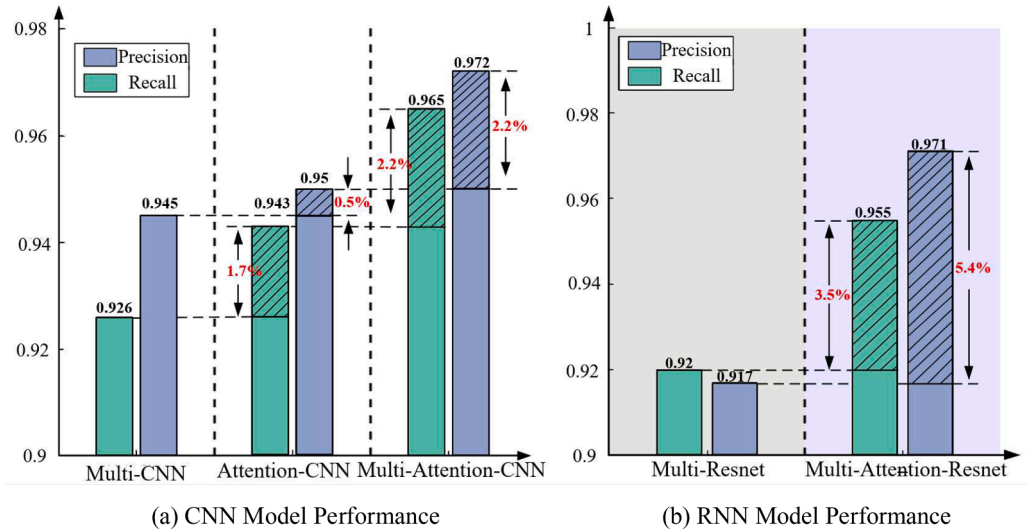


Fig. 13. Normal wear.

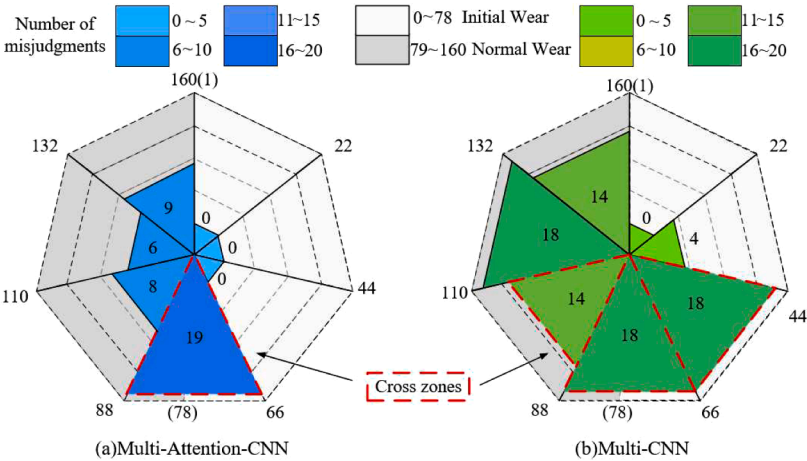


Fig. 14. Misjudgment distribution during initial and normal wear stages.

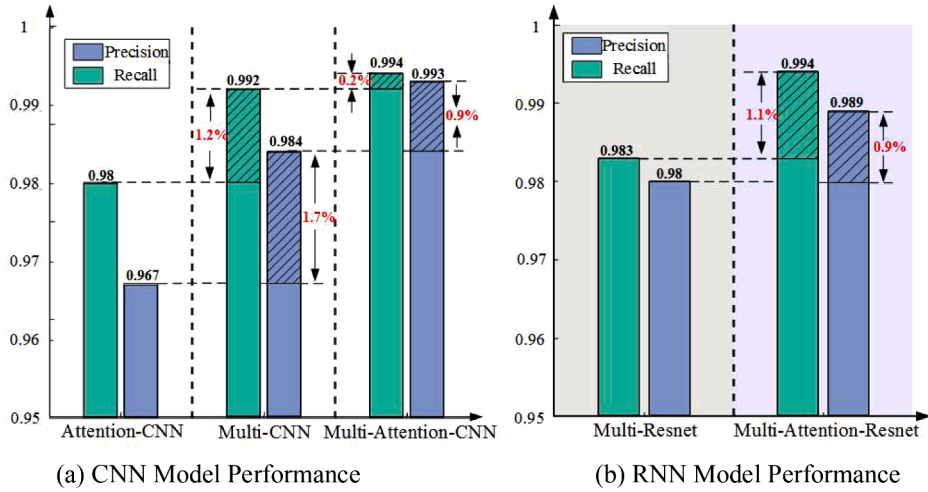


Fig. 15. Severe wear.

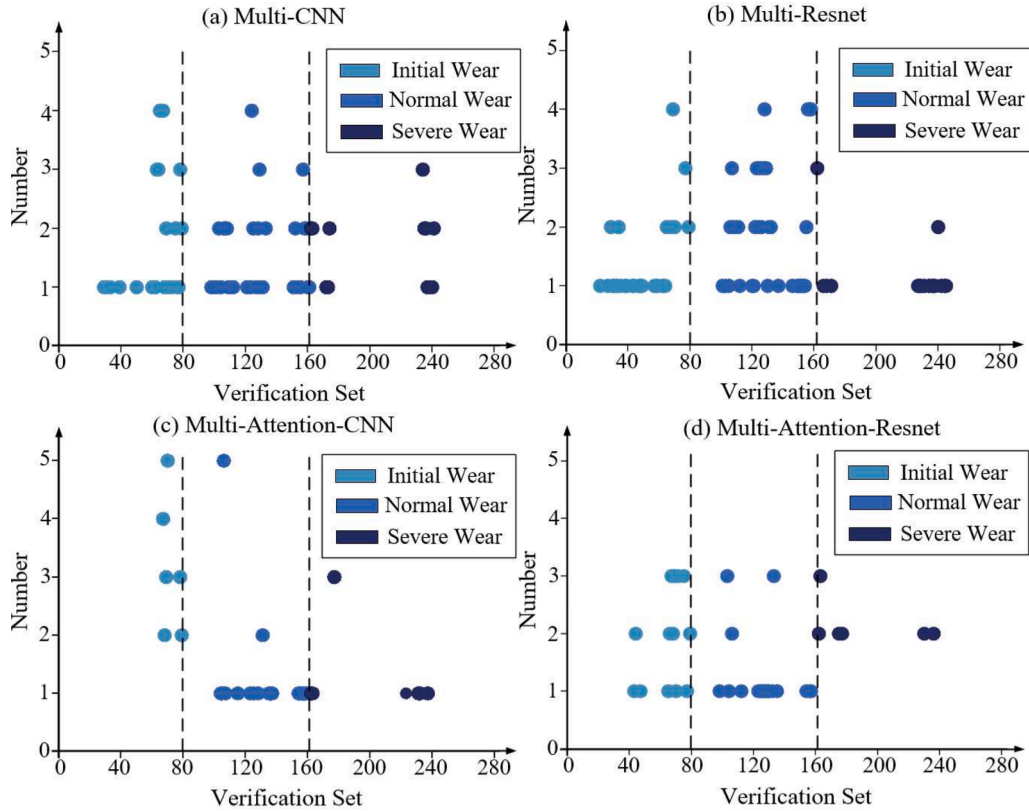


Fig. 16. Misjudgment distribution of 10 training sessions.

different cutting signals is quite distinct. Only using a single signal cannot provide comprehensive identification information for the model. Given the aforementioned issues, this work presents a novel approach for intelligent detection of tool state. The proposed method is predicated on the integration of multi-source information drive and a multi-dimensional attention mechanism. The following are the study's principal findings.

1. Comprehensive model for intelligent detection of cutting tool wear condition is developed, utilizing many sources of information. By constructing a channel-spatial mechanism for attention, different weights are given to the cutting tool wear feature information. Simultaneously, constructing multi-source information fusion is undertaken. This avoids the limitation of a single signal and achieves the complementarity of

wear information. This research does tool state identification using the PHM2010 dataset. The total identification rate of the Multi-Attention-CNN model is 0.982, and the F1 score of the initial, normal and severe wear stages of cutting tool wear is 0.977, 0.968 and 0.993, which are better than other models.

2. In the initial wear and normal wear stage, the main reason for misjudgment is the cross area between the wear stages. In order to address the aforementioned issues, this paper improves the overall performance of the model by constructing a channel-spatial mechanism for attention. The fuzzy regions of various wear stages have been reduced. The overarching performance of the model in the initial wear and normal wear stages increased by 2.6 % and 3.3 %. The accuracy and recall rate of initial wear identification reached 0.978 and 0.975. The

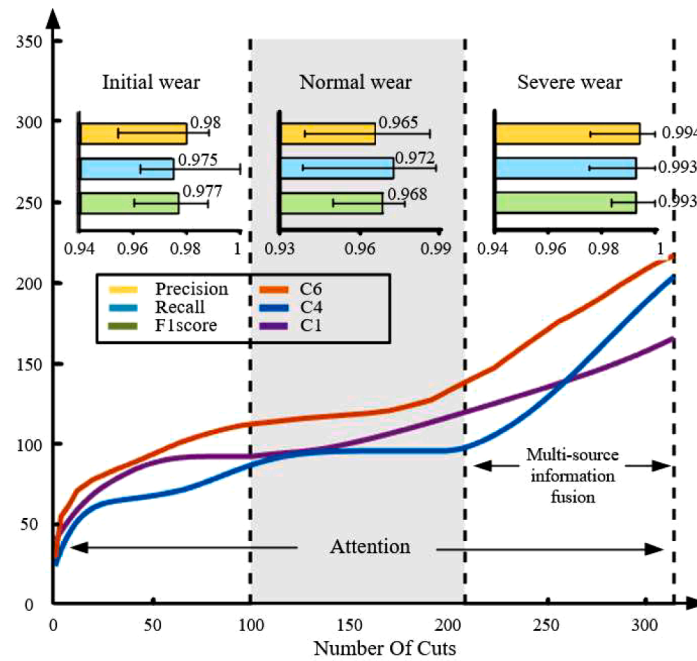


Fig. 17. Comprehensive Performance of Multi-Attention-CNN Mode.

accuracy and recall rate of normal wear identification reached 0.965 and 0.972

3. In the severe wear phase, a single signal message contains missing or redundant information elements. This will greatly affect the cutting tool state identification. Aiming at this problem, a multi-source signal fusion module is constructed. This realizes the complementarity of information in time and space, enhances the model's capacity for object perception and recognition. Under the premise of global optimization of channel-spatial mechanism for attention, multi-source signal fusion is established. This results in a precision and recall of 0.994 and 0.993 for the model in the severe wear phase, which is 1.4 % and 2.6 % higher.

The approach for tool state identification developed in this research is capable of offering dependable information for the identification of tool state. However, due to the limitation of time and cost, the basic structure of the model built in this paper has not been modified by network pruning and other methods. In the study of this paper, the original data were not processed for noise reduction and the construction of cutting tool monitoring system. Although the identification of each cutting tool wear stage of the model is studied in this paper, the reverse optimization of the model is not realized through the research results. We will continue to optimize the model. The model prediction accuracy and training efficiency are taken into account simultaneously. Redundant structures are trimmed. Model training time is reduced. According to the contribution of different modules, the local optimization of the model is realized. At the same time, the channel attention mechanism is further combined with multi-source information fusion. The tool wear surveillance system is constructed. The monitoring of tool wear values is implemented.

CRedit authorship contribution statement

Peining Wei: Writing – original draft, Methodology, Visualization. **Rongyi Li:** Writing – review & editing, Formal analysis, Software. **Xianli Liu:** Supervision, Project administration, Writing – review & editing. **Haining Gao:** Data curation. **Mingqiu Dai:** Data curation, Visualization. **Yuhan Zhang:** Data curation, Software. **Wenkai Zhao:** Investigation, Writing – review & editing. **Erliang Liu:** Validation.

Declaration of competing interest

This manuscript has not been published or presented elsewhere in part or in entirety and is not under consideration by another journal. We have read and understood your journal's policies, and we believe that neither the manuscript nor the study violates any of these. There are no conflicts of interest to declare.

Data availability

Data will be made available on request.

Acknowledgments

The authors are grateful for financial support from the Natural Science Foundation of Heilongjiang Province (No. TD2022E003), Open Fund of Key Laboratory of Advanced Manufacturing and Intelligent Technology, Ministry of Education (No. KFKT202202), National Key Research and Development Program of China (2019YFB1704800) and the National Natural Science Foundation of China (52075135).

References

- [1] A.R.F. Oliveira, L.R.R. Da Silva, V. Baldin, et al., Effect of tool wear on the surface integrity of Inconel 718 in face milling with cemented carbide tools[J], *Wear*, 476 (2021) 203752.
- [2] X. Liang, Z. Liu, B. Wang, et al., Modeling of plastic deformation induced by thermo-mechanical stresses considering tool flank wear in high-speed machining Ti-6Al-4V[J], *Int. J. Mech. Sci.* 140 (2018) 1–12.
- [3] M. Cheng, L. Jiao, P. Yan, et al., Intelligent tool wear monitoring and multi-step prediction based on deep learning model[J], *J. Manuf. Syst.* 62 (2022) 286–300.
- [4] K. Zhu, T. Liu, Online tool wear monitoring via hidden semi-Markov model with dependent durations[J], *IEEE Trans. Industr. Inform.* 14 (1) (2017) 69–78.
- [5] M. Kiouss, A. Ouahabi, M. Boudraa, et al., Detection process approach of tool wear in high speed milling[J], *Measurement* 43 (10) (2010) 1439–1446.
- [6] R. Teti, Machining of composite materials[J], *CIRP Ann.* 51 (2) (2002) 611–634.
- [7] W. Wenqi, L. Wei, Y. Zhang, et al., Precise measurement of geometric and physical quantities in cutting tools inspection and condition monitoring: a review[J], *Chin. J. Aeronaut.* (2023).
- [8] Z. Qiao, Y. Lei, N. Li, Applications of stochastic resonance to machinery fault detection: a review and tutorial[J], *Mech. Syst. Signal. Process.* 122 (2019) 502–536.

- [9] H. Wang, W. Yue, S. Wen, et al., An improved bearing fault detection strategy based on artificial bee colony algorithm[J], *CAAI. Trans. Intell. Technol.* 7 (4) (2022) 570–581.
- [10] N.N. Bhat, S. Dutta, S.K. Pal, et al., Tool condition classification in turning process using hidden Markov model based on texture analysis of machined surface images [J], *Measurement* 90 (2016) 500–509.
- [11] X. Wei, X. Liu, C. Yue, et al., Tool wear state recognition based on feature selection method with whitening variational mode decomposition[J], *Robot. Comput. Integr. Manuf.* 77 (2022) 102344.
- [12] Y. Xie, C. Zhang, Q. Liu, Tool wear status recognition and prediction model of milling cutter based on deep learning[J], *IEEe Access.* 9 (2020) 1616–1625.
- [13] J. Sun, G.S. Hong, M. Rahman, et al., Improved performance evaluation of tool condition identification by manufacturing loss consideration[J], *Int. J. Prod. Res.* 43 (6) (2005) 1185–1204.
- [14] J. Sun, G.S. Hong, M. Rahman, et al., Identification of feature set for effective tool condition monitoring by acoustic emission sensing[J], *Int. J. Prod. Res.* 42 (5) (2004) 901–918.
- [15] C. Liu, G.F. Wang, Z.M. Li, Incremental learning for online tool condition monitoring using Ellipsoid ARTMAP network model[J], *Appl. Soft. Comput.* 35 (2015) 186–198.
- [16] G. Wang, Y. Yang, Q. Xie, et al., Force based tool wear monitoring system for milling process based on relevance vector machine[J], *Adv. Eng. Softw.* 71 (2014) 46–51.
- [17] S. Dong, P. Wang, K. Abbas, A survey on deep learning and its applications[J], *Comput. Sci. Rev.* 40 (2021) 100379.
- [18] L. Alzubaidi, J. Zhang, A.J. Humaidi, et al., Review of deep learning: concepts, CNN architectures, challenges, applications, future directions[J], *J. Big. Data* 8 (2021) 1–74.
- [19] Y. LeCun, Y. Bengio, G. Hinton, Deep learning[J], *Nature* 521 (7553) (2015) 436–444.
- [20] Z. Wu, G. Luo, Z. Yang, et al., A comprehensive review on deep learning approaches in wind forecasting applications[J], *CAAI. Trans. Intell. Technol.* 7 (2) (2022) 129–143.
- [21] G. Van Houdt, C. Mosquera, G. Nápoles, A review on the long short-term memory model[J], *Artif. Intell. Rev.* 53 (2020) 5929–5955.
- [22] Z. Li, F. Liu, W. Yang, et al., A survey of convolutional neural networks: analysis, applications, and prospects[J], *IEEe Trans. Neural Netw. Learn. Syst.* (2021).
- [23] G. Serin, B. Sener, A.M. Ozbayoglu, et al., Review of tool condition monitoring in machining and opportunities for deep learning[J], *Int. J. Adv. Manuf. Technol.* 109 (2020) 953–974.
- [24] M. Ma, C. Sun, X. Chen, et al., A deep coupled network for health state assessment of cutting tools based on fusion of multisensory signals[J], *IEEe Trans. Industr. Inform.* 15 (12) (2019) 6415–6424.
- [25] W. Li, H. Fu, Z. Han, et al., Intelligent tool wear prediction based on Informer encoder and stacked bidirectional gated recurrent unit[J], *Robot. Comput. Integr. Manuf.* 77 (2022) 102368.
- [26] X.C. Cao, B.Q. Chen, B. Yao, et al., Combining translation-invariant wavelet frames and convolutional neural network for intelligent tool wear state identification[J], *Comput. Ind.* 106 (2019) 71–84.
- [27] H. Sun, J. Zhang, R. Mo, et al., In-process tool condition forecasting based on a deep learning method[J], *Robot. Comput. Integr. Manuf.* 64 (2020) 101924.
- [28] X. Xu, Z. Tao, W. Ming, et al., Intelligent monitoring and diagnostics using a novel integrated model based on deep learning and multi-sensor feature fusion[J], *Measurement* 165 (2020) 108086.
- [29] J. Ou, H. Li, G. Huang, et al., Tool wear recognition based on deep kernel autoencoder with multichannel signals fusion[J], *IEEe Trans. Instrum. Meas.* 70 (2021) 1–9.
- [30] J. Ou, H. Li, G. Huang, et al., Intelligent analysis of tool wear state using stacked denoising autoencoder with online sequential-extreme learning machine[J], *Measurement* 167 (2021) 108153.
- [31] X. Li, X. Liu, C. Yue, et al., A data-driven approach for tool wear recognition and quantitative prediction based on radar map feature fusion[J], *Measurement* 185 (2021) 110072.
- [32] J. Zhou, C. Yue, X. Liu, et al., Classification of tool wear state based on dual attention mechanism network[J], *Robot. Comput. Integr. Manuf.* 83 (2023) 102575.
- [33] T. Mikolajczyk, K. Nowicki, A. Kłodowski, et al., Neural network approach for automatic image analysis of cutting edge wear[J], *Mech. Syst. Signal. Process.* 88 (2017) 100–110.
- [34] R. Geirhos, J.H. Jacobsen, C. Michaelis, et al., Shortcut learning in deep neural networks[J], *Nat. Mach. Intell.* 2 (11) (2020) 665–673.
- [35] X. Xu, J. Wang, B. Zhong, et al., Deep learning-based cutting tool wear prediction and its application for machining process using multi-scale feature fusion and channel attention mechanism[J], *Measurement* 177 (2021) 109254.
- [36] M. Jaderberg, K. Simonyan, A. Zisserman, Spatial transformer networks[J], *Adv. Neural Inf. Process. Syst.* (2015) 28.
- [37] F. Wang, M. Jiang, C. Qian, et al., Residual attention network for image classification[C], *Proc. IEEE Conf. Comput. Vision Pattern Recognit.* (2017) 3156–3164.
- [38] J. Hu, L. Shen, G. Sun, Squeeze-and-excitation networks[C], *Proc. IEEE Conf. Comput. Vision Pattern Recognit.* (2018) 7132–7141.
- [39] PHM Society, PHM Society Conference Data Challenge, 2010 <https://www.phmsociety.org/competition/phm/10> 2010 (accessed 20 December 2021).
- [40] A. Siddhpura, R. Paurobally, A review of flank wear prediction methods for tool condition monitoring in a turning process[J], *Int. J. Adv. Manuf. Technol.* 65 (1) (2013) 371–393.
- [41] T. Mohanraj, S. Shankar, R. Rajasekar, et al., Tool condition monitoring techniques in milling process—a review[J], *J. Mater. Res. Technol.* 9 (1) (2020) 1032–1042.
- [42] C.H. Lauro, L.C. Brandão, D. Baldo, et al., Monitoring and processing signal applied in machining processes—a review[J], *Measurement* 58 (2014) 73–86.
- [43] M.U. Sathianarayan Gudelek, G. Serin, A.M. Ozbayoglu, et al., An industrially viable wavelet long-short term memory-deep multilayer perceptron-based approach to tool condition monitoring considering operational variability[J], *Proc. Inst. Mech. Eng., Part E* 237 (6) (2023) 2532–2546.
- [44] B. an, Y.C. Singh Samant, P.S. Conjeevaram Guruprasad, et al., Feature-based augmentation and classification for tabular data[J], *CAAI. Trans. Intell. Technol.* 7 (3) (2022) 481–491.
- [45] M.H. Guo, T.X. Xu, J.J. Liu, et al., Attention mechanisms in computer vision: a survey[J], *Comput. Vis. Media (Beijing)* 8 (3) (2022) 331–368.
- [46] A. Vaswani, N. Shazeer, N. Parmar, et al., Attention is all you need[J], *Adv. Neural Inf. Process. Syst.* (2017) 30.
- [47] X. Li, W. Wang, X. Hu, et al., Selective kernel networks[C], *Proc. IEEE/CVF Conf. Comput. Vision Pattern Recognit.* (2019) 510–519.
- [48] S. Woo, J. Park, J.Y. Lee, et al., Cbam: convolutional block attention module[C]//, *Proc. Eur. Conf. Comput. Vision (ECCV)* (2018) 3–19.